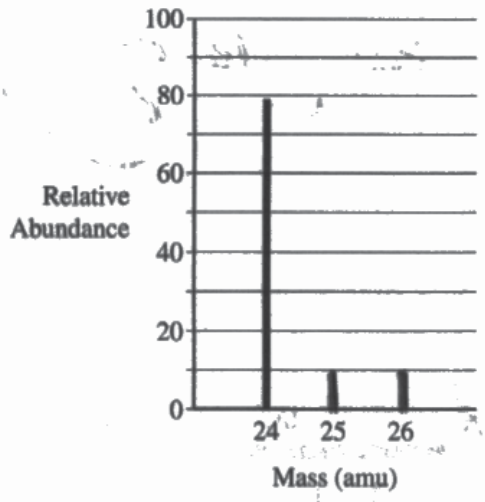
Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Part A

Question 1



ii) Magnesium-25 has a +1 ionic charge making it's atomic mass less than that of Mg-24. Similarly, Mg-26 has a smaller mass than Mg-25 & Mg-24, because of a +2 ion. These ions cause the atom to go left a period and therefore lose electrons.

Question 1 is continued on the next page.

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Question 1

Part B ✓

Coulombs law: Magnitude of the attractive force varies directly with the product of opposite charges & indirectly w/ savare the distance btw. attractive forces.

i) The charge of the Na^+ ion has a weaker attraction to water than Mg^{2+} bcuz Mg is ~~dosen't~~ more electro negative.

ii) Mg^{+2} is larger because it has a bigger electron shell.

Part C ✓

$pH = -\log [H_3O^+]$

$pH = -\log [2.80 \times 10^{-4}]$

$pH = 3.553$

Question 1 is continued on the next page.

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Question 1

Part D ✓

$Mg(NO_3)_{2\,(aq)} + NaOH_{(aq)} \rightarrow NO_3Na + (OH)_2Mg$

$[Mg^{2+}] = 28.57$

molarity = $\frac{moles}{L} = \frac{1}{0.035} = 28.57$

Part E ✓

i) $K_{sp} = \frac{[Mg][OH^2]}{[Mg(OH)^2]}$

ii) COMMENT $Q = 5.5 \times 10^{-5}$ $\frac{1.65 \times 10^{-4}}{3}$

iii) A precipitate will form because $Q > K$, and therefore there will be sufficient energy to create a precipitate.

Question 1 is continued on the next page.

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Question 1

Part F ✓

Since the student added a strong acid, $HNO_{3(aq)}$, the amount of undissolved $Mg(OH)_{2(s)}$ increases because as a strong acid HNO_3 ionize completely in aqueous solutions, And therefore reacts w/ $Mg(OH)_2$ and Creates more product.

Continue to Question 2.